

Chapter 4, Section 1

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1. Let X be a curve, and let $P \in X$ be a point. Then there exists a nonconstant rational function $f \in K(X)$, which is regular everywhere except at P .

Proof. For large enough $n \gg 0$, $\ell(nP) > 1$ by (1.3.2). This implies the existence of an effective divisor D' such that $D' - nP = (f)$, where f is a rational function and (f) is the divisor of zeros of f . Since D' is effective, the only pole of f is P of order at most n . $\square$

5. For an effective divisor D on a curve X of genus g , show that $\dim |D| \leq \deg D$. Furthermore, equality holds if and only if $D = 0$ or $g = 0$.

Proof. The Riemann-Roch theorem gives

$$\ell(D) - 1 = \ell(K - D) + \deg D - g,$$

where $\dim |D| = \ell(D) - 1$, so it suffices to show $\ell(K - D) \leq g$. If D is effective, we have an exact sequence

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_D \longrightarrow 0,$$

and we can tensor by $\mathcal{O}_X(K)$ to get

$$0 \longrightarrow \mathcal{O}_X(K - D) \longrightarrow \mathcal{O}_X(K) \longrightarrow \mathcal{O}_D \longrightarrow 0.$$

Taking cohomology gives an injection $H^0(\mathcal{O}_X(K - D)) \rightarrow H^0(\mathcal{O}_X(K))$, so we conclude that $\ell(K - D) \leq \ell(K) = g$.

Now, we have shown $\dim |D| = \deg D$ if and only if $\ell(K - D) = \ell(K)$, and it is clear this equality holds if either $D = 0$ or $g = 0$. Conversely, suppose $D \neq 0$ and $g \neq 0$. We will show $\ell(K - D) < \ell(K)$. Since D is effective, $d = \deg D > 0$, so we proceed by induction on d . If $D = P$ for a point $P \in X$, then $\ell(P) = 1$ because X is not rational. Thus, Riemann-Roch gives

$$\ell(K - P) = g - 1 < g = \ell(K).$$

Now, assume the statement for $d > 1$, and let D be an effective divisor of degree $d + 1$. Let P be a point in the support of D so that $D' = D - P$ is an effective divisor of degree d . By the inductive hypothesis, we have $\ell(K - D') < \ell(K)$. Consider the exact sequence

$$0 \longrightarrow \mathcal{O}_X(-P) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_P \longrightarrow 0.$$

Tensoring with $\mathcal{O}_X(K - D')$, we get

$$0 \longrightarrow \mathcal{O}_X(K - D) \longrightarrow \mathcal{O}_X(K - D') \longrightarrow \mathcal{O}_P \longrightarrow 0,$$

so there is an inclusion $H^0(\mathcal{O}_X(K - D)) \hookrightarrow H^0(\mathcal{O}_X(K - D'))$. Thus, $\ell(K - D) \leq \ell(K - D')$, so combining inequalities, we get

$$\ell(K - D) \leq \ell(K - D') < \ell(K).$$

$\square$

6. Let X be a curve of genus g . Show that there is a finite morphism $f : X \rightarrow \mathbf{P}^1$ of degree $\leq g + 1$.

Proof. Fix a point $P \in X$, and consider the divisor $D = (g + 1)P$. By Riemann-Roch, $\ell(D) > 1$, which implies the existence of a nonconstant rational function $f \in K(X)$ with poles no worse than D . This means $f = g/h$, where g and h are global functions on X of same degree such that the order of vanishing of h at P is at most $g + 1$, and h has no other zeros. The inclusion of fields $k(f) \subseteq K(X)$ induces a finite morphism $f : X \rightarrow \mathbf{P}^1$. Now, the fibre of f at $0 \in \mathbf{P}^1$ is at most $g + 1$ points, so f has degree at most $g + 1$ by (II, 6.9). $\square$

7. A curve X is called *hyperelliptic* if $g \geq 2$ and there exists a finite morphism $f : X \rightarrow \mathbf{P}^1$ of degree 2.

- (a) If X is a curve of genus $g = 2$, show that the canonical divisor defines a complete linear system $|K|$ of degree 2 and dimension 1, without basepoints. Use (II, 7.8.1) to conclude that X is hyperelliptic.
- (b) Show that the curves constructed in (1.1.1) all admit a morphism of degree 2 to $\mathbf{P}^1$. Thus, there exist hyperelliptic curves of any genus $g \geq 2$.

Proof.

- (a) By Riemann-Roch, $\deg K = 2g - 2 = 2$ and $\dim |K| = \ell(K) - 1 = 1$. For any $P \in X$, consider the exact sequence

$$0 \longrightarrow \mathcal{O}_X(K - P) \longrightarrow \mathcal{O}_X(K) \longrightarrow \mathcal{O}_P \longrightarrow 0.$$

Then $|K|$ is basepoint free if and only if $H^0(\mathcal{O}_X(K)) \rightarrow H^0(\mathcal{O}_P)$ is surjective for every $P \in X$. Taking the long exact sequence in cohomology gives

$$0 \longrightarrow H^0(\mathcal{O}_X(K - P)) \longrightarrow H^0(\mathcal{O}_X(K)) \longrightarrow H^0(\mathcal{O}_P),$$

and $h^0(\mathcal{O}_X(K)) = 2$ and $h^0(\mathcal{O}_P) = 1$, so it suffices to show $h^0(\mathcal{O}_X(K - P)) = 1$. Indeed, Riemann-Roch gives $\ell(K - P) = \ell(P)$, and since X is not rational, $\ell(P) = 1$. Hence, $\ell(K - P) = 1$.

- (b) Let X be an irreducible and nonsingular divisor of type $(g + 1, 2)$ on $\mathbf{P}^1 \times \mathbf{P}^1$ given by a closed embedding $j : X \rightarrow \mathbf{P}^1 \times \mathbf{P}^1$, and let $\pi : \mathbf{P}^1 \times \mathbf{P}^1 \rightarrow \mathbf{P}^1$ be the projection map onto its first argument. We claim $f = \pi \circ j : X \rightarrow \mathbf{P}^1$ is a finite morphism of degree 2. For any $P \in \mathbf{P}^1$, $\pi^{-1}(P)$ is a divisor of type $(1, 0)$, and the intersection of X and $\pi^{-1}(P)$ will generally be 2 distinct points. Hence, f has degree 2 by (II, 6.9).

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